VitalFlow V1 — Claims dashboard + Denial queue + Balances

Day-to-day surface for billers, office admins, and practice owners. Turns the charge + claim + denial + balance data from billing-rcm.md into a work-ready dashboard.

See also:

- docs/billing-rcm.md — underlying domain and state machines
- docs/charge-capture.md — where charges originate
- docs/permissions-matrix.md — `billing:*`

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1. Pages and routes

All under `apps/web/src/app/(app)/billing/`. Every page is a Server Component; state lives in the URL (querystring-driven filters + pagination).

```
/billing                              → redirects to /billing/claims
├── /billing/claims                   Claims list (filters + pagination)
├── /billing/claims/[id]              Claim detail
├── /billing/denials                  Denial work queue
├── /billing/denials/[id]             Denial detail
└── /billing/balances                 Patient balances (aging dashboard)
```

Not in V1: `/billing/payments`, `/billing/remittance`.

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2. Table columns

Claims list — 9 columns: Claim #, Patient, Payer, Status, Service dates, Total, Paid, Pat Resp, Last activity. Default sort `updated_at desc`.

Denial queue — Priority (★1–5), Claim #, Codes (chips), Amount, Age (days), Status, Assignee, Created. Default sort `priority asc, created_at asc`; default filter `status in ('open','working')`.

Balances — Patient, Current, 0–30, 31–60, 61–90, 90+, Last payment, Last statement. Default sort `aging_over_90_minor desc`; over-90 > 0 cells rendered red.

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3. Filters

URL-driven (`<form method="GET">` , no client JS).

- **Claims:** status (multi), payer, provider, service date from/to, free-text q, page
- **Denials:** status (multi, default open+working), assignee (me/unassigned/any/specific), priority, free-text code, page
- **Balances:** aging band (all/0-30/31-60/61-90/90+), min balance, free-text patient search, page

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4. Status badges

Single `<StatusBadge>` primitive mapping enum to semantic tone:

Status	Tone
Claims: draft, closed	muted
Claims: submitted, partial, appealed	warning
Claims: ready, accepted	info
Claims: paid	success
Claims: rejected, denied	destructive
Denials: open	destructive
Denials: working, appealed	warning
Denials: resolved	success
Denials: written_off, uncollectable	muted

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5. Detail layouts

Claim detail — header (claim #, status badge, totals) → two-column (patient+payer+provider / perm-gated action buttons) → claim lines table → linked denials mini-list → status history timeline.

Denial detail — header (codes, amount, priority, age, status) → claim context + assignment panel → reason → append-only work log → action panel (Resolve, Appeal, Write off).

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6. Action model

All writes are Server Actions with POST-redirect-GET (matches encounter workspace pattern). Every action: session + permission check → admin SQL write → `revalidatePath` → `redirect` with `?ok=` or `?error=`.

Action	File	Perm	Audit event
`markClaimReady`	claims/actions.ts	billing:write	`claim.status_changed`
`submitClaim`	same	billing:write	banner "clearinghouse not wired" (501-equiv)
`applyRemittance`	same	billing:write	`claim.status_changed` + per-line `denial.created`
`appealClaim`	same	billing:write	`claim.appealed`
`closeClaim`	same	billing:write	`claim.status_changed`
`assignDenial` / `assignToMe`	denials/actions.ts	billing:write	`denial.assigned`
`recordDenialWork`	same	billing:write	row-level audit only
`resolveDenial`	same	billing:write	`denial.resolved`
`writeOffDenial`	same	billing:write_off	`write_off.applied` + `denial.resolved`
`appealDenial`	same	billing:write	status change via row-level audit

`submitClaim` + `applyRemittance` are the two **clearinghouse-bound** actions that return a "not wired" banner in V1. Everything else is fully functional.

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7. Permissions

Surface	Required
`/billing/**`	`billing:read` (layout gate)
Mutating claim actions	`billing:write`
Write-off	`billing:write_off`
Top-nav "Billing" link	`billing:read`

Physicians, nurses, schedulers: no access. Billers, office admins, practice owners: full access. Impersonation strips `billing:adjust` and `billing:write_off` (write-off buttons disabled for impersonators).

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8. Empty / loading / error states

Empty — copy tuned to filter context:

- Claims (no filters): "No claims yet. Create one from Charge Capture on an encounter."
- Claims (filters): "No claims match these filters. [Clear]"
- Denials (default): "No open denials. Nicely done."
- Denials (custom): "No denials match these filters."
- Balances: "No patients with outstanding balances."

Loading — no skeleton UI in V1 (SSR + `force-dynamic` renders inline). Browser navigation bar handles transitions.

Error — two layers: per-page `?error=…` banner from failed actions; default Next.js `error.tsx` for unhandled server errors (V1 acceptable).

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9. Acceptance criteria

- [ ] `/billing` redirects to `/billing/claims`.
- [ ] User with `billing:read` sees all four pages; user without is 403-redirected from the layout.
- [ ] Claims list renders with default sort `updated_at desc`.
- [ ] Claim status filter applies via querystring; URL is shareable.
- [ ] Claim detail shows header, patient, payer, lines, status-history timeline.
- [ ] `Mark ready` button visible only when `status=draft` + `billing:write`; action transitions to `ready` and inserts history row.
- [ ] `Submit` returns a 501-equivalent banner (clearinghouse not wired).
- [ ] `Appeal` form requires reason ≥ 5 chars; transitions status; emits `claim.appealed`.
- [ ] Denial queue default shows `open` + `working`, sorted priority asc + created asc.
- [ ] Age column shows days since created; red when > 30.
- [ ] `Assign to me` sets `assigned_to` and `assigned_at`; emits `denial.assigned`.
- [ ] `Record work` is append-only (adds a dated block to `work_note`).
- [ ] `Resolve` transitions status; emits `denial.resolved`.
- [ ] `Write off` refused without `billing:write_off`.
- [ ] Balances page sorts by `aging_over_90_minor desc` by default; 90+ cells red when > 0.
- [ ] All filter forms are `<form method="GET">` — no client JS.
- [ ] Monorepo typecheck passes; no `"use client"` introduced.

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10. Not V1

- Payments list + record-payment UI (service exists in 5A; UI deferred).

- Real 835 upload + parse (clearinghouse seam only).
- Statement generation (prerequisite for the `last_statement_at` column).
- Payment-plan / autopay.
- Collections workflow (dunning).
- Denial rule engine + auto-assignment.
- CSV export.
- Cross-tenant dashboard (super-admin).

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11. Overview dashboard

`/billing` is the overview. "Overview" is the first tab; the old redirect to `/billing/claims` is gone.

11.1 Layout

```
┌─ filter bar: date range + provider ──────────────────┐
├─ KPI row:  Charges posted · Open denials · Patient A/R · Claims in range │
├─ Panels row 1: Claims by status  |  Aging snapshot │
└─ Panels row 2: Recent payments   |  Denial priority breakdown │
```

Two columns at ≥ md, stacked at sm. All Server Components; the filter bar is a `<form method="GET">`.

11.2 KPIs + sources

#	KPI	Source table	Filter-aware
1	Charges posted (count + total)	`public.charges` (status ≠ voided, service_date in range)	yes
2	Open denials (count + amount + priority≤2 + age>30)	`public.denials` (status in open/working)	no (point-in-time)
3	Patient A/R (sum + count)	`public.patient_balances` (current > 0)	no (cumulative)
4	Claims in range (count)	`public.claims` (service_start_date in range)	yes
5	Claims by status (row 2)	same as #4, grouped by status	yes
6	Aging snapshot (row 2)	sums of aging buckets on `public.patient_balances`	no
7	Recent payments (row 3)	top 8 from `public.payments` joined to patient + payer	yes
8	Denial priority breakdown (row 3)	grouped by priority, status in open/working	no

All eight queries run in parallel. Each panel's fetcher is fail-soft — one bad query doesn't crash the dashboard.

11.3 Filters

Date range (`from` , `to`) + provider (`provider`). Presets via `?range=today|7d|30d|mtd` . Defaults to today.

Flow metrics (charges, claims, payments) respect filters. State metrics (denials, A/R, aging, priority) are always point-in-time.

11.4 Drill-downs

- Status bars → `/billing/claims?status=X&from=&to=&provider=`
- Aging buckets → `/billing/balances?band=X`
- "Open denials" card → `/billing/denials`
- "Patient A/R" card → `/billing/balances`
- Recent payments → *(no payments list UI yet — follow-up)*

11.5 Acceptance

- [ ] `/billing` renders the overview (no redirect).
- [ ] Four KPI cards render with accurate counts for seeded data.
- [ ] Status bar click → `/billing/claims?status=X` with preserved range + provider.
- [ ] Aging bucket click → `/billing/balances?band=X` .
- [ ] Recent payments shows up to 8 rows sorted by received_at desc.
- [ ] Zero-data tenant renders cleanly with empty-state copy per panel.
- [ ] Filter is a `<form method="GET">` (no JS).
- [ ] Typecheck + build pass; no `"use client"` introduced.